
Analyst-Consensus ETF Rotation Strategy

A point-in-time research framework for turning stock-level analyst revisions into ETF signals, portfolio decisions, and risk-controlled leverage rules.

Core design: revision-driven signals, freshness decay, historical ETF weights, breadth and coverage checks, confounder controls, separate bearish testing, and staged validation before leverage.

RESEARCH SUMMARY | POINT-IN-TIME DESIGN | AUGUST 2026

This document presents a research hypothesis, not an established source of alpha or an investment recommendation.

Purpose

This document summarizes a proposed ETF rotation strategy that converts stock-level analyst opinion into ETF-level bullish and bearish signals. It also incorporates the main research improvements needed to make the idea testable, point-in-time correct, and suitable for eventual portfolio use.

The strategy is a research hypothesis, not an established source of alpha or an investment recommendation.

1. Strategy concept

The central idea is to:

1. collect analyst recommendations and recommendation changes for individual stocks;
2. map those stocks into ETFs using each ETF's actual constituent weights at the time;
3. calculate an ETF-level analyst signal;
4. rank or select ETFs using that signal; and
5. express the strongest positive or negative views through ordinary, leveraged, inverse, or short ETF exposure under explicit risk rules.

In compact form:

```
Point-in-time analyst revisions on stocks
      ↓
Freshness- and quality-weighted stock signals
      ↓
Point-in-time ETF constituent weights
      ↓
ETF signal + breadth + coverage filters
      ↓
Long/short ETF selection and regime-aware position sizing
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Using ETFs rather than individual stocks can diversify company-specific risk and translate clusters of analyst opinion into sector, industry, theme, or broad-market exposures. It also creates potential overlap problems, so highly similar ETFs must not be treated as independent bets.

2. Bullish workflow

1. Gather point-in-time analyst actions for stocks, including upgrades and upward changes in recommendation.
2. Convert each action into a standardized positive or negative revision score.
3. Apply a freshness decay so recent revisions receive more weight.
4. Optionally weight revisions by analyst or institution quality.
5. Map the stock signals into ETFs using historical, point-in-time constituent weights.
6. Measure whether the bullish signal is broad across holdings or concentrated in a few large stocks.
7. Require sufficient analyst-data coverage before an ETF is eligible.
8. Rank eligible ETFs by the final bullish signal.
9. Use ordinary 1x ETFs for positive signals by default. Consider leveraged ETFs only for the strongest, broadest signals and only when the market-regime and risk filters permit leverage.

3. Bearish workflow

1. Gather downgrades and downward recommendation changes at the stock level.

2. Convert them into negative revision scores and apply the same freshness and quality logic used on the bullish side.
3. Aggregate the signals through historical ETF constituent weights.
4. Check bearish breadth and analyst-data coverage.
5. Rank ETFs by the strength of their negative signal.
6. Test three implementations separately: staying out of negatively ranked ETFs, shorting ordinary ETFs, and buying inverse ETFs.
7. Consider inverse or leveraged-inverse exposure only after accounting for trading costs, borrow constraints, volatility, daily-reset compounding, and the prevailing market regime.

The bearish side should not be assumed to be the mirror image of the bullish side. Shorting costs, rebound risk, market drift, crowded positioning, and inverse-fund path dependence can make its behavior structurally different.

4. Make the signal revision-driven

The primary signal should be the **change in analyst opinion**, not simply the current recommendation level.

Examples of positive revisions include:

Hold → Buy
 Buy → Strong Buy
 Neutral → Outperform
 Underweight → Equal Weight

Examples of negative revisions include:

Strong Buy → Hold
 Buy → Sell
 Overweight → Neutral
 Equal Weight → Underweight

Recommendation levels may still be retained as a secondary feature, but the main hypothesis is that new information in upgrades and downgrades is more informative than a stale consensus level. Recommendation labels must be standardized across institutions before scoring.

One simple mapping assigns ordered numerical values to normalized ratings. For example:

Strong Sell = -2
 Sell = -1
 Hold = 0
 Buy = +1
 Strong Buy = +2

For analyst action k on stock i :

$$\text{Revision}_{i,k,t} = \text{Rating}_{i,k,t}^{\text{new}} - \text{Rating}_{i,k,t}^{\text{old}}$$

Alternative scoring systems should be tested because a one-notch change may not have the same meaning across all brokers, sectors, or starting ratings.

5. Freshness decay

Recent revisions should receive more weight than older ones. A basic exponentially decayed stock signal is:

$$\text{StockSignal}_{i,t} = \sum_{k \in K(i,t)} \text{Revision}_{i,k,t} \times \text{Quality}_{k,t} \times e^{-\lambda \text{Age}_{i,k,t}}$$

where:

- $K(i,t)$ is the set of eligible analyst actions for stock i known by time t ;

- Quality[k,t] is initially 1 and can later represent analyst quality;
- Age[i,k,t] is the age of the action in trading days; and
- λ controls the speed of decay.

For a half-life of h trading days:

$$\lambda = \ln(2) / h$$

A reasonable first test is h=20 trading days, followed by sensitivity tests such as 5, 10, 20, 40, and 60 days. The half-life should be selected through out-of-sample validation rather than chosen from the full sample.

To prevent stocks with more analyst activity from mechanically dominating, compare at least two variants:

$$\text{StockSignal}_{i,t}^{\text{sum}} = \sum_k \text{WeightedRevision}_{i,k,t}$$

and

$$\text{StockSignal}_{i,t}^{\text{avg}} = [\sum_k \text{WeightedRevision}_{i,k,t}] / [\sum_k \text{Quality}_{k,t} e^{-\lambda \text{Age}_{i,k,t}}]$$

The first captures both direction and intensity; the second captures average direction while reducing analyst-coverage bias.

6. ETF weight aggregation

Aggregate stock-level signals using the ETF's actual point-in-time constituent weights:

$$\text{ETFSignal}_{j,t}^{\text{raw}} = \sum_{i \in H(j,t)} \text{Weight}_{ij,t} \times \text{StockSignal}_{i,t}$$

where $H(j,t)$ is the set of holdings in ETF j at time t, and $\text{Weight}[ij,t]$ is stock i's ETF weight known at that time.

This is preferable to merely counting how many favorably rated stocks an ETF holds because a 10% holding should normally affect the ETF signal more than a 0.1% holding.

Historical holdings must be point-in-time accurate. Using today's constituents to reconstruct old ETF signals would introduce survivorship and look-ahead bias. The backtest must also respect the publication delay between a holdings date and when that information became available.

7. Breadth

The weighted signal can be dominated by one or two large positions. Breadth measures whether analyst revisions agree across a meaningful share of the ETF.

Useful breadth measures include:

$$\text{PositiveBreadth}_{j,t} = [\sum_i \text{Weight}_{ij,t} 1(\text{StockSignal}_{i,t} > 0)] / [\sum_i \text{Weight}_{ij,t} 1(\text{StockSignal}_{i,t} \neq 0)]$$

and the unweighted version:

$$\text{PositiveNameBreadth}_{j,t} = \#\{i : \text{StockSignal}_{i,t} > 0\} / \#\{i : \text{StockSignal}_{i,t} \neq 0\}$$

Bearish breadth is defined analogously. A signal is more convincing when both its magnitude and breadth are strong. Breadth may be used as a filter, as a separate ranking feature, or as an interaction term with the raw ETF signal.

8. Coverage

ETFs differ greatly in how much of their portfolios is covered by analysts. Define weight coverage as:

$$\text{Coverage}_{j,t} = \sum_{i \in H(j,t)} \text{Weight}_{ij,t} \times 1(\text{ValidSignal}_{i,t})$$

An optional coverage-adjusted signal is:

$$\text{ETFSignal}_{j,t}^{\text{covered}} = \text{ETFSignal}_{j,t}^{\text{raw}} / \text{Coverage}_{j,t}$$

provided coverage is above a minimum threshold. Both the raw signal and the coverage-adjusted signal should be retained: the raw measure describes total portfolio exposure, while the adjusted measure describes the average signal among covered holdings.

Coverage should never be silently treated as zero sentiment. An ETF with little analyst coverage is an uncertain observation, not necessarily a neutral one.

9. Analyst-quality extension

The first version should give each eligible analyst action equal quality weight. A later extension can estimate whether certain analysts or institutions have historically produced more useful revisions.

Possible quality inputs include:

- out-of-sample post-revision return accuracy;
- sector-specific rather than universal skill;
- recommendation timeliness relative to peers;
- forecast accuracy, where available;
- consistency across market regimes; and
- sample-size shrinkage so analysts with a few lucky calls do not receive extreme weights.

A generic quality weight can be written as:

$$\text{Quality}_{a,s,t} = f(\text{HistoricalSkill}_{a,s,<t}, \text{SampleSize}_{a,s,<t})$$

where the weight for analyst a in sector s at time t is calculated only from information available before t . Quality weights should be capped and shrunk toward 1. This extension must be validated out of sample; otherwise it creates a large overfitting surface.

10. Momentum and earnings confounder controls

Analysts often revise ratings after price moves or around earnings. A profitable raw signal could therefore be repackaged price momentum, earnings surprise, post-earnings-announcement drift, or another known effect.

At the stock level, control for at least:

- recent returns over short and medium horizons;
- distance from recent highs;
- earnings-announcement indicators;
- standardized earnings surprise;
- recent forecast revisions, if available;
- size, liquidity, volatility, value, and sector effects; and
- market and industry momentum.

At the ETF level, control for:

- ETF price momentum;
- sector or theme momentum;
- market beta and volatility;

- liquidity, spreads, and assets under management; and
- concentration and overlap with other selected ETFs.

Recommended tests include:

1. double-sort ETFs on momentum and analyst signal;
2. regress future ETF returns on the analyst signal plus momentum and earnings controls;
3. residualize the analyst signal using only data available at the time;
4. exclude a short window around earnings and compare results;
5. separate revisions that lead price moves from revisions that follow them; and
6. test whether the strategy adds value beyond a simple ETF momentum strategy.

The key question is not whether analyst revisions correlate with returns, but whether they add incremental predictive information after these confounders are controlled.

11. Market-regime leverage rules

Leverage should be a portfolio overlay, not part of the core signal. First establish that the unlevered signal works.

A simple rule set is:

- use 1x exposure for ordinary positive signals;
- permit leverage only when signal magnitude, breadth, and coverage all exceed precommitted thresholds;
- block or reduce leverage during high-volatility, poor-liquidity, or adverse-trend regimes;
- scale total exposure to a volatility target;
- cap exposure by ETF, sector, theme, and correlated cluster;
- use a drawdown-based de-risking rule; and
- rebalance often enough to keep the signal current but not so often that turnover consumes the edge.

One illustrative leverage rule is:

$$\text{Leverage}_{j,t} = \min[L_{\max}, \text{TargetVol} / \text{ForecastVol}_{j,t}] \times \text{SignalGate}_{j,t} \times \text{RegimeGate}_t$$

where SignalGate reflects signal strength, breadth, and coverage, and RegimeGate is between 0 and 1. All thresholds and regime definitions must be specified before final out-of-sample testing.

Leveraged and inverse ETFs reset daily. Their multi-day return depends on the path of daily returns, so a backtest must use the funds' actual tradable returns or accurately model daily resetting, fees, financing, and tracking error. Multiplying an ordinary ETF's long-horizon return by two or three is not an adequate simulation.

12. Separate bearish and inverse testing

Treat the following as distinct strategies:

1. **Long-only rotation:** overweight strongly positive ETFs and hold cash or a benchmark instead of negative ETFs.
2. **Long/short rotation:** long positive ETFs and short ordinary negative ETFs.
3. **Long/inverse rotation:** long positive ETFs and buy inverse ETFs corresponding to negative exposures.
4. **Leveraged long/inverse overlay:** use leveraged products only after the underlying unlevered signals pass validation.

For each variant, report gross and net performance separately. Include bid-ask spreads, commissions, borrow fees and availability, management fees, financing, tracking error, slippage, turnover, taxes where relevant, and daily-reset effects.

The bearish strategy should have its own signal thresholds, holding periods, decay parameters, and risk limits. These values may legitimately differ from the bullish side, but they must be selected without using the final test set.

13. Portfolio construction and implementation safeguards

Before ranking ETFs, define an eligible universe using investability criteria such as listing history, assets, volume, spreads, and availability of reliable historical holdings.

Additional safeguards include:

- use timestamps for recommendation release, holdings availability, prices, and trades;
- impose a realistic delay between observing information and trading;
- eliminate survivorship bias by including delisted and closed ETFs where possible;
- account for ticker, fund, index, and share-class changes;
- prevent duplicated exposure by clustering ETFs with highly overlapping holdings;
- cap concentration in any underlying stock that appears across multiple ETFs;
- avoid tuning many thresholds on one historical period; and
- report capacity and turnover, not just risk-adjusted return.

Signal standardization should preferably occur within comparable ETF groups or include category controls. A broad-market ETF, a concentrated semiconductor ETF, and a niche thematic ETF should not automatically be assumed to have identical signal distributions.

14. Staged research roadmap

Stage 1 — Data audit

- Verify point-in-time analyst timestamps and standardized rating histories.
- Verify historical ETF constituents, weights, and their actual publication times.
- Build a survivorship-bias-aware ETF universe.
- Quantify missingness, analyst coverage, and holdings frequency.

Gate: proceed only if the data can support a genuinely point-in-time test.

Stage 2 — Stock-level signal validation

- Test recommendation levels versus revisions.
- Test alternative revision encodings and decay half-lives.
- Measure returns after upgrades and downgrades across horizons.
- Control for momentum, earnings, size, liquidity, volatility, and sector.

Gate: revisions must show incremental out-of-sample information, not merely in-sample correlation.

Stage 3 — Unlevered ETF aggregation

- Aggregate stock signals using historical ETF weights.
- Compare raw, coverage-adjusted, breadth-filtered, and equal-weighted variants.
- Test ranking portfolios and signal thresholds.
- Cluster overlapping ETFs and enforce exposure limits.

Gate: the 1x ETF strategy must survive costs and reasonable alternative specifications.

Stage 4 — Robustness and falsification

- Use walk-forward or nested time-series validation.
- Reserve a final untouched test period.
- Test across market regimes, sectors, ETF types, and rebalance frequencies.
- Run placebo timestamps, shuffled signals, and delayed-signal tests.
- Compare against ETF momentum, sector rotation, and benchmark portfolios.
- Correct for multiple testing and report parameter sensitivity.

Gate: results should be stable enough that success does not depend on one narrow parameter choice or subperiod.

Stage 5 — Bearish implementation

- Test avoidance, ordinary ETF shorts, and inverse ETFs separately.
- Model borrow, daily reset, fees, spreads, and tracking error.
- Estimate downside-tail and rebound risk.

Gate: retain bearish exposure only if it improves net portfolio outcomes relative to simply reducing long exposure.

Stage 6 — Leverage overlay

- Apply volatility targeting and regime gates to a validated 1x strategy.
- Test leverage caps, drawdown controls, and concentration limits.
- Use actual leveraged-fund returns where products existed.
- Stress test gap moves, volatility spikes, and extended choppy markets.

Gate: leverage must improve the objective after costs and tail risk, not merely magnify gross return.

Stage 7 — Paper trading and monitoring

- Run the full process prospectively without capital.
- Log every input, signal, eligibility decision, and hypothetical trade.
- Compare expected and realized turnover, spreads, tracking, and signal decay.
- Predefine model-change and shutdown criteria before live deployment.

15. Core formula set

Stock-level decayed revision signal

$$\text{StockSignal}_{i,t} = \sum_k \text{Revision}_{i,k,t} \times \text{Quality}_{k,t} \times e^{-\lambda \text{Age}_{i,k,t}}$$

Decay parameter

$$\lambda = \ln(2) / \text{HalfLife}$$

ETF-level weighted signal

$$\text{ETFSignal}_{j,t}^{\text{raw}} = \sum_i \text{Weight}_{ij,t} \times \text{StockSignal}_{i,t}$$

Analyst-data coverage

$$\text{Coverage}_{j,t} = \sum_i \text{Weight}_{ij,t} \times 1(\text{ValidSignal}_{i,t})$$

Coverage-adjusted ETF signal

$$\text{ETFSignal}_{j,t}^{\text{covered}} = \text{ETFSignal}_{j,t}^{\text{raw}} / \text{Coverage}_{j,t}$$

Weighted bullish breadth

$$\text{PositiveBreadth}_{j,t} = [\sum_i \text{Weight}_{ij,t} 1(\text{StockSignal}_{i,t} > 0)] / [\sum_i \text{Weight}_{ij,t} 1(\text{StockSignal}_{i,t} \neq 0)]$$

Illustrative regime-aware leverage

$$\text{Leverage}_{j,t} = \min[L_{\max}, \text{TargetVol} / \text{ForecastVol}_{j,t}] \times \text{SignalGate}_{j,t} \times \text{RegimeGate}_t$$

16. Final research question

The strategy should ultimately answer:

Do recent stock-level analyst recommendation revisions, aggregated through point-in-time ETF constituent weights, breadth, coverage, and optionally analyst quality, predict future ETF returns after controlling for momentum, earnings-related information, common risk exposures, overlapping holdings, and realistic trading costs—and, only if they do, can regime-aware inverse or leveraged implementation improve risk-adjusted portfolio outcomes without unacceptable path dependence or tail risk?

That question should be answered first for unlevered bullish ETF rotation, then for bearish implementations, and only finally for leveraged overlays.